东正阳

4 $\Delta p = mv$

$F\Delta t = n\Delta p \qquad n = 120$

解 $F = 11.613\ N$

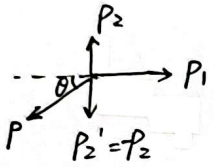
.7 $m = \rho_1 l$

$\Delta F = \dot{p} = (\rho_1 v_0 t \cdot v_0)' = \rho_1 v_0^2$

$F = mg + \Delta F$

解 $F = \rho_1 l g + \rho_1 v_0^2$

3.9



$P = \sqrt{P_1^2 + (P_2')^2} = 10.65 \times 10^{-21}\ kg \cdot m/s$

$= 1.065 \times 10^{-20}\ kg \cdot m/s.$

方向与P_1反方向夹角为θ且 $\tan\theta = \frac{533}{922} \approx 0.58$

3.14 (1) $F \cdot dt = v dm$

$dm = v' dt$ 其中 $v' = 1.29 \times 10^3\ kg/s$

$F = mg$

解 $v = 2.07 \times 10^4\ m/s$

(2) 废气全部喷出时，火箭上升速率最大

$M dv - u dm = -Mg dt$

$-dm = dM$

解 $\frac{dv}{dt} = -\frac{u}{M}\frac{dM}{dt} - g$

$v = \int dv = \int -\frac{u}{M} dM - g dt = -u \ln M - gt$

$v = \int_0^v dv = \int_M^{M-\Delta m} -\frac{u}{M} dM - \int_0^T g dt = -u \ln M \Big|_M^{M-\Delta m} - gT$

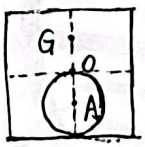
$= 2.68 \times 10^3\ m/s.$

(3) $h = \int_0^T v dt = \int_0^T \left(u \ln \frac{M}{M-kt} - gt\right) dt = u \ln M T + \frac{u}{k}(M-kT)[\ln(M-kT) - 1] - \frac{1}{2} g T^2$

$-\frac{u}{k} M (\ln M - 1)$

$= 2.03 \times 10^5\ m = 203\ km$

3.18



由对称性中，剩余铜块的质心G在截面中心竖线上

完整铜块的质心O在正方体体心处

由力矩平衡有

$m_1 L_1 = m_2 L_2$

m_1为剩余铜块的质量 $m_1 = \rho(a^3 - \frac{\pi}{16} a^3)$

m_2为圆柱形铜块的质量 $m_2 = \rho \frac{\pi}{16} a^3$

$L_2 = \frac{a}{4}$

解 $L_1 = 0.061a$.

可知剩余铜块的质心在正方体体心上方0.061a处

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3.19 前者的速度 $v_1 = gt$

后者的速度 $v_2 = g(t-t_0)$

$$v_c = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

解 $v_c = (t - \frac{5}{7})g$　方向竖直向下

$$a_c = \frac{m_1 g + m_2 g}{m_1 + m_2}$$

解 $a_c = g$　方向竖直向下

3.22 (1) 由角动量守恒定律有

$$m v_1 l_1 = m_2 v_2 l_2$$

$$v_2 = \frac{v_1 l_1}{l_2} = \frac{10.2\ km/s \cdot (205.5+6378)km}{(35835.7+6378)km} = 1.59\ km/s$$

(2) 椭圆轨道的长半轴长 $a = \frac{35835.7+205.5}{2} km + 6378\ km = 24398.6\ km$

半焦距 $c = \frac{35835.7-205.5}{2} km = 17815.1\ km$

易知短半轴长 $b = \sqrt{a^2-c^2} = 16670.7\ km$

$$L = m r v \sin\alpha = m r \left|\frac{d\vec{r}}{dt}\right| \sin\alpha = m \lim_{\Delta t \to 0} \frac{r|\Delta\vec{r}| \sin\alpha}{\Delta t}$$

又 $\Delta S = \frac{1}{2} \Delta r \sin\alpha \cdot r$　则 $L = 2m \frac{\Delta S}{\Delta t}$

$$S = \int \Delta S = \int \frac{L}{2m} \Delta t = \frac{L}{2m} T$$

又 $S = \pi ab$

则 $T = \frac{2m\pi ab}{L} = \frac{2\pi ab}{v_1 l_1}$

解 $T = 10.57\ h$

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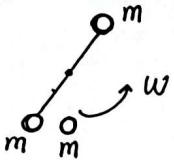
3.24 (1)



显然质心一定在硬上

$2m\,l_1 = m\,l_2$

$l_1 + l_2 = a$

解 $l_1 = \frac{a}{3}$

即质心在~~靠近~~将粘在一起的质点的三等分点

$V_c = \frac{mv + m(-v) + 0}{3m} = 0$

即质心速度为0

(2) $L = mv\cdot\frac{a}{3} + mv\cdot\frac{2}{3}a + 0 = mva = \frac{1}{2}mwa^2$

碰后一瞬间，对于系统而言，无外力作用，故角动量守恒

即碰后一瞬间角动量仍为 $\frac{1}{2}mwa^2$

(3) $2m\cdot\frac{a}{3}w'\cdot\frac{a}{3} + m\cdot\frac{2}{3}aw'\cdot\frac{2}{3}a = L$

解 $w' = \frac{3}{4}w$

即碰撞后，整个系统绕质心角速度为 $\frac{3}{4}w$.

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